

Producing algorithms: pseudocode

Revise 360

Worksheet for the 360° experience: Pseudocode workshop

Name: _____ Class: _____ Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Pseudocode workshop. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Write clear pseudocode.
- Translate between algorithm representations.
- Distinguish assignment and comparison.

Starter: before you put the headset on

What happens after $\text{score} = 5$ followed by $\text{score} = \text{score} + 2$?

.....

.....

Key terminology

Pseudocode

.....

.....

Assignment

.....

.....

The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Readable instructions

Turn right

Key facts in your own words:

.....

Example to trace:

INPUT quantity
INPUT price
total = quantity * price
OUTPUT total

Challenge: Why must pseudocode still be precise?

.....

Which is an effective pseudocode instruction?

.....

Pseudocode should be...

.....

2 Assignment or equality

Turn right

Key facts in your own words:

Example to trace:

score = 7
Stores 7 in score

score == 7
Tests whether score is 7

Challenge: Explain the different jobs of = and == here.

Match the expression to its role.

Match each item to a description from the list below.

Greater-than comparison / Equality comparison / Assignment

Item	Matching description
count = 0	
count == 0	
count > 0	

After x = 4 then x = 7, x contains...

3 Write selection

Turn around

Key facts in your own words:

Example to trace:

```
if age >= 12 then
    print("Allowed")
else
    print("Too young")
endif
```

Challenge: What output is produced for age 12?

Which line belongs at the end in OCR notation?

For age 11, the displayed selection outputs...

4 Write repetition

Turn around

Key facts in your own words:

Example to trace:

```
total = 0
for i = 1 to 3
    value = int(input())
    total = total + value
next i; print(total)
```

Challenge: Why is print(total) after the loop?

Order the running-total design.

Put these steps in order: Output total / Finish the loop / Repeat input and addition three times / Initialise total to 0

Why initialise total once?

5 Translate to Python

Turn left

Key facts in your own words:

Example to trace:

```
# Python
for i in range(1, 4):
    print(i)
```

Outputs: 1, 2, 3

Challenge: Why is the stop value 4 rather than 3?

Python range(3) produces which values?

Python range(1, 3) yields...

6 Check equivalence

Turn left

Key facts in your own words:

Example to trace:

OCR: for i = 1 to 3
Python: range(1, 4)
Both repeat 3 times
Check the actual values too

Challenge: Why can a direct copy of loop bounds be wrong?

What best checks equivalent behaviour?

Equivalent representations should produce...

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

OCR for `i = 2 to 4` prints `i`. Which Python range matches?

Which describes pseudocode?

Your score

Copy it from the score screen as soon as it appears.

_____ / 19

Key question

How can you explain an algorithm without choosing a particular programming language?

Review and improve

Use review mode. Explain one correction and why the correct answer fits. Write the three-number total in Python, run it with 3, 4 and 5, and compare the output with your pseudocode. Expected total: 12.

Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Write pseudocode to input quantity and unit price and output the total. [3]

b) Write selection that outputs Free if cost == 0 and Pay otherwise. [3]

c) Explain the loop-bound difference between OCR for and Python range. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Write clear pseudocode.			
Translate between algorithm representations.			
Distinguish assignment and comparison.			